

Classical Edge Detector Ablation Study

Comprehensive Parameter Sweep Across Noise Regimes

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1. Motivation

Prior experiments compared the WVF/LF against four deep learning edge detectors (DexiNed, TEED, pidinet, NBED) and found that the WVF with appropriately tuned parameters beats every DL model at all tested noise levels below $\text{SNR} \approx 4$. However, the study omitted classical edge detectors beyond the basic Sobel operator. Several classical methods—particularly the Canny pipeline, Laplacian of Gaussian, and Gaussian derivative filters at multiple scales—are more natural comparisons for the WVF because they share its core principle: convolution-based gradient estimation with scale as a tunable parameter.

This study adds 10 classical edge detectors to the comparison, each swept across its parameter space at the same 8 SNR levels and 5 noise types used in the existing ablation. The goal is to determine whether the WVF’s advantage over DL models extends to well-tuned classical methods, or whether simpler operators with appropriate scale parameters already achieve comparable performance.

2. Methods and Parameter Grids

2.1. Gradient Operators (1st Derivative)

2.1.1. Sobel

The Sobel operator computes horizontal and vertical image gradients using separable convolution kernels. The standard 3×3 kernel is widely used but its fixed scale makes it noise-sensitive. Extended Sobel operators at larger kernel sizes are equivalent to Gaussian derivative filters and provide a direct comparison to the WVF’s multi-scale support.

Parameter	Values	Count
Kernel size k	3, 5, 7, 9, 11, 15, 21, 31, 41, 51	10

Table 1: Sobel parameter grid. Kernel size k gives an effective support of $\approx \pi(k/2)^2$ pixels. Sobel-9 $\approx$ WVF $N_p = 64$; Sobel-31 $\approx N_p = 755$; Sobel-51 $\approx N_p = 2043$, well beyond the WVF’s maximum of 500.

2.1.2. Prewitt

Identical structure to Sobel but with equal-weight kernels (no center-weighted smoothing). Included to test whether the Sobel weighting scheme matters under noise.

Parameter	Values	Count
Kernel size k	3, 5, 7, 9, 11, 15, 21, 31, 41	9

Table 2: Prewitt parameter grid. Extended to match Sobel’s spatial range.

2.1.3. Scharr

The Scharr operator uses optimized 3×3 coefficients for improved rotational symmetry compared to Sobel. It has no scale parameter—only one configuration is tested.

Parameter	Values	Count
Kernel size k	3 (fixed)	1

Table 3: Scharr parameter grid.

2.1.4. Roberts Cross

The Roberts Cross operator uses 2×2 diagonal difference kernels. It is the smallest possible edge detector and provides a lower bound on performance at every noise level.

Parameter	Values	Count
Kernel size k	2 (fixed)	1

Table 4: Roberts Cross parameter grid.

2.2. 2nd Derivative Methods

2.2.1. Laplacian of Gaussian (LoG)

The LoG convolves the image with the Laplacian of a Gaussian kernel, producing zero-crossings at edges. The Gaussian σ controls the scale. Edge maps are obtained by detecting zero-crossings in the LoG response and thresholding the gradient magnitude at the crossing points.

Parameter	Values	Count
σ (Gaussian scale)	0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0, 8.0, 12.0, 16.0	10

Table 5: LoG parameter grid. $\sigma = 4 \approx \text{WVF } N_p = 200$ (effective radius $\approx 3\sigma = 12\text{px}$). $\sigma = 16$ far exceeds the WVF range.

2.2.2. Difference of Gaussians (DoG)

The DoG approximates the LoG by subtracting two Gaussian-blurred images at different scales. The ratio $k = \sigma_2/\sigma_1$ controls the bandwidth.

Parameter	Values	Count
σ_1	0.5, 1.0, 2.0, 3.0, 4.0, 6.0, 8.0, 12.0	8
$k = \sigma_2/\sigma_1$	1.6 (standard)	1

Table 6: DoG parameter grid. $\sigma_2 = k \cdot \sigma_1$.

2.3. Compass / Directional Operators

2.3.1. Kirsch

The Kirsch compass operator applies 8 directional 3×3 kernels and takes the maximum response. It provides an explicit multi-orientation search similar in spirit to the WVF’s orientation sweep, but at fixed 3×3 scale.

Parameter	Values	Count
Directions	8 (fixed, 45° spacing)	1

Table 7: Kirsch parameter grid. Single configuration — 8 fixed 3×3 compass masks.

2.3.2. Robinson

Similar to Kirsch but with different kernel coefficients (based on the Sobel template rotated to 8 directions). Provides a second fixed-scale directional comparison.

2.3.3. Frei-Chen

The Frei-Chen edge detector projects the local 3×3 neighborhood onto an orthogonal basis of 9 masks, then measures the projection onto the 4-dimensional edge subspace. It separates edge, line, and average components analytically.

Parameter	Values	Count
Basis size	9 masks (4 edge, fixed)	1

Table 8: Frei-Chen parameter grid. Single configuration using the standard 9-mask orthogonal basis.

2.4. Multi-Step Pipelines

2.4.1. Canny

The Canny edge detector applies Gaussian smoothing, Sobel gradient computation, non-maximum suppression, and hysteresis thresholding. Two parameters are swept: the Gaussian σ and the hysteresis threshold ratio. The low threshold is set as a fraction of the high threshold, which is determined by the gradient magnitude distribution.

Parameter	Values	Count
σ (Gaussian smoothing)	0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0, 8.0, 12.0, 16.0	10
Low/high threshold ratio	0.3, 0.4, 0.5	3

Table 9: Canny parameter grid. 30 configurations total. The high threshold is set at the 90th percentile of the NMS output; the low threshold is the high threshold multiplied by the ratio. $\sigma = 16$ produces extremely blurred gradients, matching a WVF with $N_p > 1000$.

Note: Canny produces binary edge maps, not continuous gradient magnitudes. ODS evaluation sweeps thresholds on the hysteresis output, which is already binary—effectively testing only the single threshold defined by the parameters. To make Canny comparable to the other methods, we also evaluate the NMS output (before hysteresis) as a continuous map, sweeping thresholds in the standard way.

2.4.2. Marr-Hildreth

The Marr-Hildreth detector applies LoG followed by zero-crossing detection. It produces binary edges and is parameterized by the LoG σ .

Parameter	Values	Count
σ	0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0, 8.0, 12.0, 16.0	10

Table 10: Marr-Hildreth parameter grid.

2.5. Scale-Space / Steerable Methods

2.5.1. Gaussian Derivatives

First-order Gaussian derivative filters at arbitrary scale. This is the continuous-scale generalization of Sobel and the most direct classical analogue to the WVF: both compute directional derivatives at a controllable spatial scale. The key difference is that Gaussian derivatives use isotropic smoothing while the WVF uses polynomial fitting with orientation-specific support.

Parameter	Values	Count
σ (Gaussian scale)	0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0, 8.0, 12.0, 16.0, 20.0	11

Table 11: Gaussian derivative parameter grid. The WVF at $N_p = n$ has effective radius $r \approx \sqrt{n/\pi}$; the comparable Gaussian $\sigma \approx r/2$. At $N_p = 50$, $\sigma \approx 2.0$; at $N_p = 250$, $\sigma \approx 4.5$; at $N_p = 500$, $\sigma \approx$

6.3. Values of $\sigma = 12$ –20 exceed the WVF range, testing whether even larger support helps.

2.5.2. Steerable Filters

Steerable filters (Freeman and Adelson, 1991) compute oriented gradient responses by analytically rotating a basis set of filters. They enable continuous orientation estimation without discrete sweeping—unlike the WVF which tests N_s discrete angles.

Parameter	Values	Count
σ (Gaussian scale)	1.0, 2.0, 3.0, 4.0, 6.0, 8.0, 12.0, 16.0	8
Order	1st derivative, 2nd derivative	2

Table 12: Steerable filter parameter grid. 16 configurations total. $\sigma = 16$ gives an effective diameter of ≈ 96 pixels, far exceeding the WVF’s largest support.

3. Summary of Configurations

Method	Configs	Parameters	Type
Sobel (extended)	10	$k \in \{3...51\}$	1st derivative
Prewitt (extended)	9	$k \in \{3...41\}$	1st derivative
Scharr	1	fixed 3×3	1st derivative
Roberts Cross	1	fixed 2×2	1st derivative
Kirsch	1	8 compass masks	Directional
Robinson	1	8 compass masks	Directional
Frei-Chen	1	9 basis masks	Subspace
Laplacian of Gaussian	10	$\sigma \in \{0.5...16\}$	2nd derivative
Difference of Gaussians	8	$\sigma_1 \in \{0.5...12\}$	2nd derivative
Canny NMS	10	$\sigma \in \{0.5...16\}$	Pipeline
Gaussian derivatives	11	$\sigma \in \{0.5...20\}$	Scale-space
Steerable 2nd order	8	$\sigma \in \{1...16\}$	Scale-space
Total classical	71		
WVF (from prior ablation)	462	$N_p \times N_s \times d$	Polynomial fit
LF (from prior ablation)	112	$N_p \times N_s \times m$	Line averaging
DL models	5	fixed weights	Learned
Grand total	650		

Table 13: Complete configuration count. Classical methods add 93 configurations per image per noise condition. Combined with existing WVF/LF and DL results, the full comparison spans 672 method-parameter combinations.

4. Scale Equivalence Reference

To compare classical methods with the WVF at matched spatial scale, Table 14 maps WVF support sizes to equivalent classical filter parameters. The effective support of the WVF is a circle of radius $r = \sqrt{N_p/\pi}$; a Gaussian derivative filter with σ has 99% of its energy within radius 3σ ; a Sobel kernel of size k covers a square of side k .

WVF N_p	Eff. radius	Eff. diameter	Equiv. Gaussian σ	Equiv. Sobel k
15	2.2 px	4 px	0.7	5
25	2.8 px	6 px	0.9	5
50	4.0 px	8 px	1.3	7
100	5.6 px	11 px	1.9	11
250	8.9 px	18 px	3.0	17
500	12.6 px	25 px	4.2	25
LF $m = 7$	—	15 px (line)	2.5	15
LF $m = 14$	—	29 px (line)	4.8	29
LF $m = 20$	—	41 px (line)	6.8	41

Table 14: Scale equivalence between WVF/LF support and classical filter parameters. “Equiv. Gaussian σ ” is computed as $r/3$ to match 99% energy. “Equiv. Sobel k ” is $\approx 2r$. The classical sweep extends to Sobel-51 ($\approx N_p = 2043$) and Gaussian $\sigma = 20$ ($\approx N_p = 11\{, \}310$), well beyond the WVF’s maximum.

5. Noise Conditions

The same noise conditions from the existing WVF/LF and DL ablation studies are used, ensuring direct comparability.

Noise Type	Model	Relevance
Gaussian	$I' = I + \mathcal{N}(0, \sigma^2)$	Sensor thermal noise (baseline)
Salt & pepper	$P(\text{pixel} = 0 \text{ or } 255) \propto \frac{1}{1+\text{SNR}}$	Dead pixels, bit errors
Poisson	$I' = \frac{\text{Pois}(I \cdot s)}{s}$	Low-light shot noise
Speckle	$I' = I \cdot (1 + \mathcal{N}(0, \frac{1}{\text{SNR}^2}))$	Sonar, radar, underwater
Uniform	$I' = I + \mathcal{U}(-a, a)$	ADC quantization noise

Table 15: Five noise types at 8 SNR levels (0.3, 0.5, 1.0, 1.5, 2.0, 3.0, 4.0, ∞).

6. Evaluation Protocol

Identical to prior ablation studies: 1001 thresholds, 3-pixel match radius, dataset-wide ODS, no post-processing. Classical methods that produce continuous gradient magnitudes (Sobel, Prewitt, Gaussian derivatives, etc.) are evaluated by threshold sweep. Methods that produce binary outputs (Canny, Marr-Hildreth) are evaluated both as binary maps and—where applicable—as the continuous intermediate representation (e.g., Canny’s NMS output before hysteresis).

7. Computational Budget

All classical edge detectors are implemented as GPU-accelerated convolutions via PyTorch’s `torch.nn.functional.conv2d`. Even the largest kernels (Sobel-51, Gaussian $\sigma = 20$) run in <5 ms per image on an A100. The full sweep of 71 configurations on one 720×1280 image completes in <0.5 seconds.

Component	Configs	Images $\times$ conditions	Est. time
Classical filters (GPU)	71	$20 \times 36 = 720$	≈ 6 min
ODS evaluation (CPU)	71	720	≈ 15 min
Total			≈ 21 min

Table 16: Estimated runtime for the classical ablation on one A100 GPU. All convolutions are GPU-accelerated via PyTorch. ODS evaluation uses the vectorized CPU pipeline.

8. Expected Outcomes

Hypothesis 1: Gaussian derivatives at matched scale will match WVF performance.

The WVF at $N_p = 50$ has an effective support radius of $r \approx 4$ pixels. A Gaussian derivative filter at $\sigma \approx 2$ uses a comparable spatial scale with isotropic smoothing. If the WVF’s advantage is primarily due to its larger support (noise averaging), then Gaussian derivatives at $\sigma = 2\text{--}4$ should perform similarly. If the WVF still wins, the advantage is in the polynomial model or orientation-specific fitting.

Hypothesis 2: Extended Sobel with large kernels will approach WVF at low SNR.

Sobel- 31×31 uses 961 pixels per gradient estimate, comparable to WVF at $N_p = 300$. At low SNR where noise averaging dominates, these should converge.

Hypothesis 3: Canny will underperform raw gradient magnitudes. Our earlier Canny NMS test showed that NMS uniformly degrades ODS by 0.03–0.14. This should hold for all classical methods: the continuous gradient magnitude outperforms the thinned/thresholded version under the ODS metric.

9. Relationship to WVF

The most important comparison is between the WVF and Gaussian derivatives at matched scale. The WVF fits a 4th-order polynomial to N_p neighbors at each orientation, extracting the normal derivative. A Gaussian derivative filter convolves with the derivative of a Gaussian, producing an equivalent directional derivative with isotropic support. The WVF has three potential advantages:

1. **Orientation-specific support:** the WVF evaluates the polynomial along the edge normal, while Gaussian derivatives use circular support. This should matter for anisotropic features.
2. **Higher-order fitting:** the WVF fits a 4th-order surface, potentially capturing curvature information that 1st-order Gaussian derivatives miss.
3. **Non-uniform neighbor weighting:** the WVF’s least-squares fit implicitly weights neighbors by their position in the Taylor expansion, while Gaussian derivatives use symmetric Gaussian weighting.

If Gaussian derivatives at $\sigma = 2\text{--}4$ match WVF at $N_p = 50\text{--}250$, then advantages 1–3 are negligible and the WVF reduces to an expensive reimplementaion of a standard technique. If the WVF consistently outperforms, the polynomial fitting provides genuine value beyond scale selection.